

Title

Resolving Heterogeneous Biological Systems One Component at a Time: From Viral Evolution to Membrane Biophysics

Abstract

Predictive models of complex biological systems require single-component, high-resolution measurements in their native context. We develop droplet microfluidic and multiplexed imaging platforms that generate high-throughput, high-content molecular measurements across biological scales from viral particles to whole-brain circuits. Applying single-virus genomic methods to influenza, we work toward resolving reassortment events at single-particle resolution, a dimension of genomic diversity critical for modeling zoonotic transmission dynamics yet inaccessible to conventional sequencing. Using double-emulsion microfluidics, we produce monodisperse giant unilamellar vesicles as controlled model membranes, establishing experimental infrastructure for quantitative biophysical modeling of membrane systems. These platforms share a common goal: generating ground-truth, single-component measurements that constrain and enable predictive models of heterogeneous biological behavior.